

RESEARCH SUMMARY AND INTERESTS

- My previous research explores the interplay between cryptography and distributed computing from both directions. It applies cryptographic techniques to improve the complexity and security of distributed protocols, and leverages distributed techniques to enhance the scalability and decentralization of secure cryptographic primitives. Overall, my work aims to organically integrate these two fields to design provably secure and efficient distributed cryptographic protocols.
- My current research interests lie in theoretical cryptography, particularly in secure protocols, including multi-party computation (MPC), secret sharing, threshold cryptography, Byzantine agreement, and zero-knowledge proofs (ZKP).

EDUCATION

- **Beijing Institute of Technology** Sept. 2023 - Jun. 2026
M.S. in Cyberspace Science and Technology (Advisor: [Haibin Zhang](#) and [Sisi Duan](#)) Beijing, China
 - GPA: 90.46/100
- **Qinghai University** Sept. 2019 - Jun. 2023
Bachelor of Engineering in Computer Science and Technology Xining, Qinghai, China
 - GPA: 4.2/5.0 (89.81/100), Rank: 2nd/176

PUBLICATIONS AND PATENTS

- [1] **Randomized vs. Deterministic? Practical Randomized Synchronous BFT in Expected Constant Time** SRDS 2025
Xufeng Zhang, Baohan Huang, Sisi Duan, Haibin Zhang [\[DOI\]](#) [\[eprint\]](#)
IEEE Symposium on Reliable Distributed Systems (SRDS; annual since 1981)
- [2] **Enhancing Permissioned Blockchains with Controlled Data Authorization** ACISP 2024
Qichang Liu, Xufeng Zhang, Sisi Duan, Haibin Zhang [\[DOI\]](#)
- [3] **System and Method for Distributed Agreement** Chinese Patent
Xufeng Zhang, Zijian Zhang, Liehuang Zhu

RESEARCH EXPERIENCE

- **ARiMS, Beijing Institute of Technology** Jan. 2024 - Oct. 2025
Supervised by Dr. [Haibin Zhang](#) and Dr. [Sisi Duan](#) Beijing, China
 - **Paper:** Randomized vs. Deterministic? Practical Randomized Synchronous BFT in Expected Constant Time (*First Author; SRDS 2025, a long-standing conference on distributed computing*)
 - Designed synchronous binary Byzantine agreement (BA) and multi-valued Byzantine agreement (MBA) protocols against strongly rushing adaptive adversaries, achieving expected $O(1)$ time and $O(n^2)$ messages with fewer communication steps than prior protocols.
 - Proposed a randomized synchronous BFT consensus paradigm with expected $O(1)$ time, $O(n)$ amortized message complexity per block proposal, and near-optimal 2Δ concrete latency under failure-free scenarios.
 - Implemented and evaluated a practical synchronous BFT system, demonstrating lower latency and higher throughput than existing protocols.
- **ARiMS, Beijing Institute of Technology** Jun. 2023 - Jan. 2024
Supervised by Dr. [Haibin Zhang](#) and Dr. [Sisi Duan](#) Beijing, China
 - **Paper:** Enhancing Permissioned Blockchains with Controlled Data Authorization (*ACISP 2024*)
 - Proposed threshold encryption with controlled authorization, a new primitive combining threshold encryption and threshold proxy re-encryption.
 - Decoupled data encryption from later authorization of decryption abilities, achieving collusion-resistant CPA security before authorization even if all authorization servers collude.
 - Enabled confidentiality with fine-grained access control for permissioned blockchains.

- **ARiMS, Beijing Institute of Technology**
Supervised by Dr. [Haibin Zhang](#) and Prof. [Shengli Liu](#)
 - **Competition:** 2024 “Financial Cryptography Cup” Innovation Competition
 - Developed asynchronous distributed key generation (ADKG) and asynchronous distributed key refreshment (ADKR) protocols with expected $O(1)$ time, supporting high thresholds and field-element keys.
 - Applied ADKG and ADKR to a distributed secure digital wallet for e-cash, providing proactive security, confidentiality, and no trusted setup.
 - **Dept. of IT, Yangtze Delta Region Institute of Tsinghua University**
Supervised by Dr. [Haibin Zhang](#)
 - Assisted in building *Tianshu*, a trusted data-space platform for secure and reliable data sharing among mutually distrustful entities.
 - Developed consensus mechanisms and the distributed fine-grained access-control and encryption module.
 - **School of CS, Qinghai University**
Supervised by Dr. [Rujia Li](#) and Prof. Yong Xie
 - **Bachelor’s Thesis:** Research and Design on Accountable Decryption Based on Intel SGX
 - Designed a TEE-based distributed accountable decryption system, ensuring that each decryption leaves a tamper-resistant audit record.

Aug. 2024 - Nov. 2024

Beijing, China

Oct. 2024 - Jan. 2025

Remote

Oct. 2022 - Mar. 2023

Xining, Qinghai, China
- PARTICIPATED RESEARCH PROGRAM
- **Cryptographic Technology for Distributed Blockchain Environments**
The National Key R&D Program of China
 - Participated in a major national research funding program in China.
 - Conducted research on distributed cryptographic protocols for secure and reliable blockchains, including Byzantine agreement, DKG, and threshold encryption.
 - This program supported my work published at IEEE SRDS 2025 and ACISP 2024, as well as the project awarded at the 2024 Financial Cryptography Cup.

Nov. 2022 - Oct. 2025

Participant

HONORS AND AWARDS

- **Postgraduate Student Scholarship**
Awarded by Beijing Institute of Technology
 - **Outstanding Student Scholarship**
Awarded by Qinghai University
 - **Third Group Prize of 2024 “Financial Cryptography Cup” Innovation Competition**
Awarded by the Central Bank of China
 - “Financial Cryptography Cup” is a national-level financial cryptography competition hosted by the Digital Currency Research Institute of the Central Bank of China.
 - **Second Group Prize in the First Gansu Provincial University Competition on Blockchain Technology Application for Talent Employment and Entrepreneurship**
Awarded by the Education Department of Gansu Province, China

2023 - 2026

(awarded three times)

2020 - 2022

(awarded two times)

Nov. 2024

Jun. 2021

SKILLS

- **Research Areas:** Cryptography, Secure Multi-Party Computation, Distributed Computing, Byzantine Agreement & Consensus, Threshold Cryptography, Zero-Knowledge Proofs, Provable Security, Blockchains
 - **Mathematical Background:** Abstract Algebra, Discrete Mathematics, Probability Theory, Linear Algebra
 - **Programming Languages & Tools:** Golang, Python, C/C++, Java, LaTeX, Git, AWS Cloud Deployment, MATLAB

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